

A Report on
HEAVY DUTY VEHICLE PLATOONING- CFD STUDY TOWARDS
ACHIEVING FUEL EFFICIENCY

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ABSTRACT

Truck Platooning is a new CAV technology which aims to improve the fuel efficiency and to increase the capacity on roads. This project evaluates the reason for this improved fuel efficiency which is the increased cooling due to the airflow and the obvious aerodynamic drag. And using Computational Fluid Dynamics (CFD) to analyze the airflow through the engine bay and the cooling effects of the airflow under various following distances for the trailing truck. The CFD is completed on the basis of the semi-truck designs created which were similar to the real-life trucks used in the Platooning Experiment conducted using Class 8 Vehicles. ^[1] This study deals with analyzing the conditions under which these losses can be reduced through Airflow analysis by monitoring the air flowing through the engine bay. This study is also conducted to validate the results of the experiment mentioned above. The analysis was completed by using the Solidworks 18 and the Ansys 19.0 Academic editions.

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LIST OF SYMBOLS

Symbol	Page
1. m : mass flow rate of the coolant (in kg/s).....	10
2. ΔT : change in temperature (in K)	10
3. Q_{gen} : Heat rate at the engine (coolant) (in kg/s).....	10
4. Q_{air} : Heat rate of air(heat generated) (in kg/s)	10
5. C_p : Specific heat capacity (coolant) (in kJ/kg.K)	10

CHAPTER 1

INTRODUCTION

HDV Platooning is defined as the linking of two or more Heavy Vehicles in convoy (Lead truck followed by the trailing or following trucks), using connectivity and automated driving support systems. ^[6] These vehicles, in our case, semi-trucks follow closely one after the other equipped with these state-of-the-art systems. These trucks maintain a previously set close distance between each other when they are connected such as in highways. This technology has great potential and scope for the transport sector. In the United States, vehicle automated solutions provider Peloton has been widely engaged in trials to show the benefits of truck platooning. In Michigan on December 2017, Peloton conducted a live demonstration of driver-assistive truck platooning. The demo featured two Peloton-equipped Volvo VNL670 class 8 trucks platooning with separation of approximately 65 feet. A total of 9 U.S. jurisdictions have already authorized automated vehicle platooning, according to a study by the Competitive Enterprise Institute. The states that have already given a green light to truck platooning are Alabama, Indiana, Kentucky, Louisiana, Mississippi, Nevada, Oregon, Utah, and Wisconsin. Seven states including Alabama, Arkansas, Mississippi, North Carolina, Kentucky, South Carolina and Texas have already taken some action to authorize platooning, while still enacting or maintaining certain restrictions. Why is this technology important? With the trucks that are automated following close one after the other and with zero reaction time, this technology can improve the safety drastically. And moreover, one can also see the cost savings especially in terms of fuel consumptions when this technology is enacted with trucks following at close proximities. The major reason for this reduced fuel consumption is the reduced aerodynamic drag on the trucks that are following the Lead Truck. But as we know, most of the good things come at a cost, here also the improved fuel efficiency in the following trucks due to the aerodynamic drag comes at a cost. At various following distances, the aerodynamic drag experienced by the following truck changes. However, a loss that goes unnoticed is the losses that occur due to the reduced airflow through the engine bay which brings about an increase in fuel consumption due to the fan-on conditions.

In order to model these losses, a study is conducted where the air flow through the engine bay is monitored. This study will be conducted using Ansys 19.0 and Solidworks 18 where Solidworks

will primarily serve as the design platform for the trucks and Ansys will be the analysis platform for this work.

Experiment Setting:

The experiment being validated here was conducted using two Class 8 Tractor Trailer combinations over a range of speeds, following distances and vehicle weights. The two trucks used were of the same make and model and therefore has similar dimensions. The Tractor-Trailer models were tested on an 8.5-mile oval testing track over a range of highway truck speeds, following distances, and weights representative of public on-road driving. The Heavy Heavy-Duty Diesel Truck Schedule was used as a starting point for this test. The fuel consumption data of the two platooned vehicles was documented against a control vehicle under the above-mentioned conditions. The speeds used here ranged from 55 mph to 70 mph whereas the following distances ranged from 20ft to 75 ft while the vehicle weights were varied between 65,000 lbs. and 80,000 lbs. The experiment analyzed the fuel consumption under all these different conditions. The results from this experiment show that the best savings for the platoon team were observed when the vehicles were travelling at a speed of 55 mph, following distance of 30ft and vehicle weight of 65,000 lbs.

CFD Study:

A prior knowledge of how aerodynamics works and how it is affected by various factors is significant in this study. The focus of this study is to validate the results obtained from the experiment above. Also, this study will be focusing on just the trailing truck and the following distance because the basic study involved here is based on the airflow over the engine bay and this factor remains the same for the lead tractor regardless of the following distance. The airflow rate is only being changed in the trailing truck due to the following distance between the lead truck and the following truck.

Beyond the Scope:

So, it is only logical to construct a situation where these losses are low enough and where the aerodynamic drag is also comparatively low enough such that the aerodynamic gains due to this situation outweigh the losses due to reduced airflow incurred in such a situation. So, the actual question that needs to be addressed here is that whether there is an optimum condition where there can be a compromise between the reduced aerodynamic drag and increased airflow so that we can

achieve the best fuel efficiency in Truck Platooning. But such a thorough analysis is beyond the scope of this project.

Before going into details, there are certain shortcomings that restrict the analysis capabilities of this project. A section on shortcomings will be included in the Chapter 3 on Methodology. And the findings of the project within these constraints will be discussed in Chapter 4 along with some of the inferences and the recommendations to move this technology forward.

CHAPTER 2

LITERATURE REVIEW AND STATE OF THE ART

1. Gillieron P and D'Hondt M ^[2] analyzed the airflow rates through the engine bay and through the heat exchangers. A model connecting the aerodynamic drag and the air flow rate was also established. The analysis also showed the various configurations where there is low drag and at the same time has high cooling through the engine bay. Such an analysis is useful for the scope of this study as it proves the importance of numerical simulations in studying the flow rates and the cooling as the experiments can be helpful in giving the necessary drag values and in measuring the drag variations.
2. Lammert M, Duran A, Diaz J, Burton K and Nicholson A ^[1] studied the fuel consumption results of two Class 8 tractor-trailer combinations platooned together compared to their standalone fuel consumption. SAE approved experiments conducted on a track across a variety of Gross Vehicle Weights, Speeds, and following distances were studied to obtain the optimum condition for fuel efficiency. This project validates the results from this experiment (Appendix A).
3. Liang K, Martensson J and Johansson KH ^[4] studied the 'platooning rate' of around 2000 heavy-duty vehicles by analyzing the vehicle position data from a region in Europe every day. They wanted to study how coordinated the platooning formation has to be to yield the best results. They found out that coordination can increase the platooning rate and the consequent fuel savings by a factor of 10 with certain adjustments to the driving schedule they tested. They concluded that the platooning rate and the fuel savings can be higher if there is much better flexibility. This project sets up a framework that can be used for future study as its evident that Automated platooning with an appropriate logic and a decision controller can provide good fuel efficiency and that the following distance need not be fixed to obtain optimum efficiencies.
4. Qichen Deng ^[8] in his paper studied about the impacts of HDV platooning on the traffic flow. He has developed a simulation framework where these studies can be made possible. He studied how platooning affects the traffic flow and how the traffic flow in turn affects the platoon formation. His observations (Appendix B) are helpful for this project as it

shows how the platoon formation will be affected when applied to a more realistic scenarios which needs to be tested experimentally. He has observed that platoon formation has a positive impact on the traffic flow. He has detailed that the increasing percentage of HDV platooning in traffic flow results in dramatic improvements on traffic efficiency while at the same time maintaining traffic safety. As for the impact of traffic on the platoon formation, he observed in his simulation that deceleration of the lead vehicle to a low speed during platoon formation will increase the platoon formation time to a large extent when tested in medium and heavy traffic conditions. These findings justify the recommendations made in Chapter 5 of this report.

CHAPTER 3

METHODOLOGY

There are numerous steps when approaching a CFD study. The first thing is to develop the model to be analyzed and do a background study and conduct some initial baseline analysis which should be built upon to prepare the final simulation. Then, decide your simulation framework by a detailed study and then decide your study method along with the parameters for the simulation. All these should be done keeping in mind the constraints of the project and the final project deliverables. All these portions will be covered in this chapter.

Geometry:

There are numerous steps involved in Computational Fluid Dynamics (CFD). The first step is the design of the geometry. The design used in this study are based on the real truck models that were available and based on this experiment. These CAD models were obtained from previous research conducted by students. The information about the equivalence the CAD models have with the real models is mentioned below.

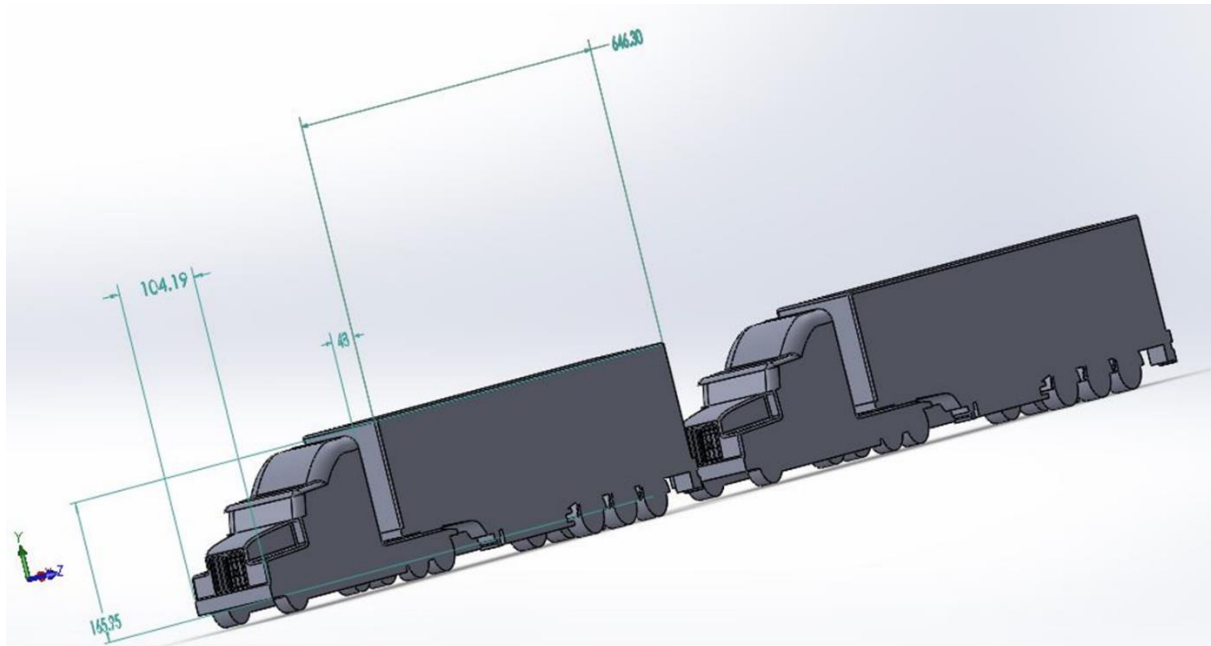


Fig 3.1: The CAD Models of the platoon formation (dimensions in inches)

The only variable in this study is the distance between the two trucks as seen in Fig 3.1. The different values used for these variables ranged from 20ft to 75ft. There are a number of possible values of this, but this project deals with the validation of this experiment and only deals with the values mentioned in the experiment. Table 3.1 shown below shows that the design specifications of the CAD models which are similar to the actual full-scale tractor-trailer models used in this experiment. All the dimensions mentioned below are also labelled in the CAD models shown in Fig 3.1.

Specifications	Lead Tractor-Trailer	Trailing Tractor-Trailer
Cab Fairing to nose of trailer	48"	48"
Trailer	53 ft Trailer	53 ft Trailer
Trailer Height	13'6"	13'6"
Trailer Width	104"	104"
Rear of side-skirt to front of trailer wheels	6.5"	8.5"

Table 3.1: Full-scale model dimensions

The table 3.1 shows that the Lead Tractor-Trailer and the Trailing Tractor-Trailer are the same and has the same design specifications. The only variable that is different here is the distance from the rear of side-skirt to front of trailer wheels for the Lead and the following trailer. This difference is alright for the purpose of our analysis since the aerodynamic effects of this gap between the side-skirt and the trailer wheels is not being considered for the following tractor as this project is focusing only on the airflow through the engine bay. There are also some other assumptions that are to be taken for this project regarding the geometry of the tractor-trailer model. Those assumptions will be explained in detail in the assumptions section in this chapter.

Since the truck is symmetrical on both sides, a half section of the trailer cut in the middle can also be taken to save computational cost. The results produced from such a model is also the same as the one you get from a complete model. Another aspect should also be considered while aiming for reduced computational cost. While using half section cuts the overall size by half, it still takes up a significant amount of computing power to analyze the half section we have. So, another step that is usually taken is to scale the model. Here, there is scaling the model by 0.01x with respect to the co-ordinate system in Solidworks which brings it within the computational limits of the

academic license of Ansys without compromising the system and putting it into Thermal Throttling. The scaled version of the trailer which has been created by scaling the model by 0.01 with respect to the origin is shown in figure 3.2.

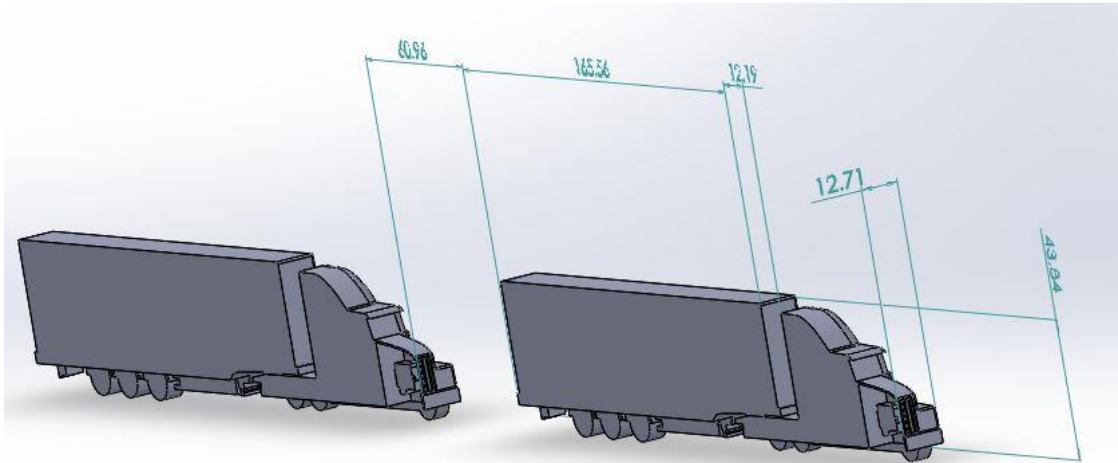


Fig 3.2: Half section of the tractor-trailer scaled by 0.01x (dimensions in mm)

However, there was a problem with using such a scaled model, which would be explained clearly in the section on Limitations. So, it is logical to revert to the full-scale model, but in order to ensure best meshing and accurate results, an inflation was added to the engine to inflate the mesh at the engine. This actually made the mesh on all other portions of the truck extremely coarse. But this was alright, since we are only focused on looking at the airflow through the engine bay.

Since the focus was on the airflow through the engine bay and because of the restricted computational availability, an alternative geometry for doing the CFD was also considered. The alternative geometry was comprised of just the engine and a box enclosing the engine. Also, an equivalent rectangle placed in front of the box can simulate the airflow draft which will be produced when the truck is in front. The only geometry that's going to affect the airflow in front of the engine bay is the back of the truck which can be replicated by the rectangular body that is mentioned before and since the airflow enters a steady state when the air is flowing over the long trailer. An equivalent rectangular body with a good enough extruded section can replicate this condition of air flow over the trailer and then through the engine bay at different following distances.

Preliminary Study:

This study considers the minor losses that is observed as a result of the increased fuel consumption due to the fan-on conditions in the engine bay. There is some ambiguity in this. Traditionally, one considers the fuel efficiency loss due to the aerodynamic drag as the major loss. However, there exists a catch, which loss is higher? Is the loss due to the aerodynamic drag more significant that we can overrule the minor loss that we see due to the increased fan usage in cases of less airflow through the engine bay?

Anyone who is familiar with vehicle aerodynamics and slipstreaming will be able to judge that the vehicle (truck in this case) following another truck closely will provide better fuel efficiency when you purely consider the efficiency loss due to the aerodynamic drag. However, as mentioned before, there is something missing here. How about the airflow through the engine bay? Anyone with a similar knowledge will also be able to judge that the airflow through the engine bay will be reduced when the truck is following closely behind another truck in comparison with a case where the truck is following at a higher distance from the front truck. Fig 3.3 can shed more light on how this becomes possible.



Fig 3.3: Airflow over the vehicle and through the front of the vehicle [10]

Fig 3.3 shows two side by side cases where one vehicle closely follows another one and another vehicle running parallel to the vehicles. It is obvious from the figure that in the 1st case the air flows over the vehicle and the airflow through the front is reduced whereas in the 2nd case the airflow through the front is higher because of uninterrupted airflow and no draft.

This effectively proves our hypothesis that the airflow through the front is reduced during closer following distance between trucks whereas the airflow increases when the following distance is increased.

A preliminary flow analysis was conducted using Solidworks Flow simulation to show that the airflow through the bay is higher when the following distance is lower. Figure 3.4 and 3.5 show two typical cases that were studied in the experiment. ^[1]

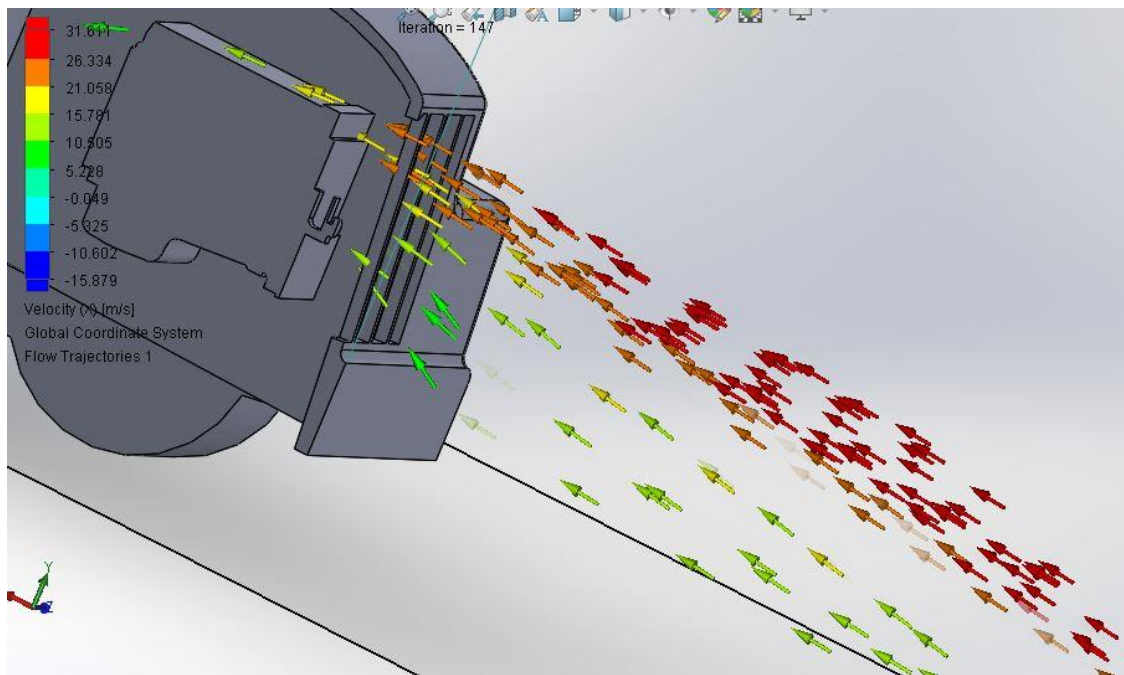


Fig 3.4: Airflow when the following distance is 20ft

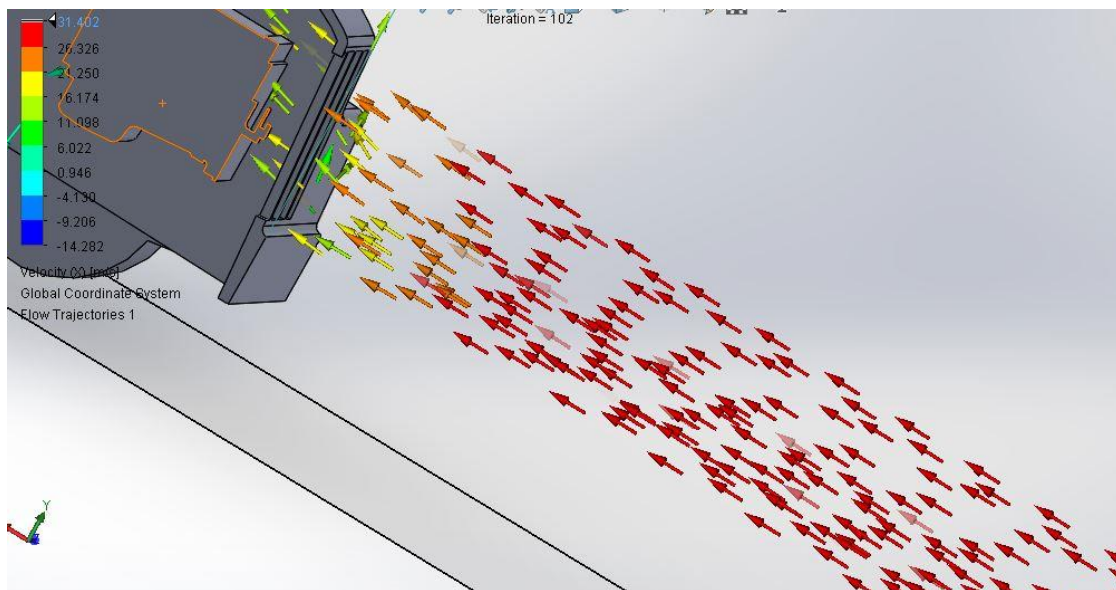


Fig 3.5: Airflow when the following distance is 75ft

Figures 3.4 and 3.5 show the air striking the front of the truck (i.e., through engine bay). It is obvious from the figures that Fig 3.5 shows more air particles striking the front of the truck (when the following distance between the two trucks is 75ft) in comparison with Fig 3.4 where fewer air particles strike the front of the truck (following distance is 20ft). This shows that when a study of airflow through the engine bay is being conducted, the favorable scenario is always the higher following distance scenario even though the higher following distance is not preferred when considering aerodynamics.

Detailed Study:

We have preliminary results which indicate that the airflow through the engine bay is higher at higher following distances and lower at lower following distances. Now, we need to validate these results above properly. There are numerous ways of doing this. An obvious way is to monitor the air flowing through the engine bay in the different cases. However, in order to show that the airflow is sufficient enough to reduce the fan running conditions, another approach is necessary. It should be proved that the natural air flow is cooling the vehicle in such a way that the engine fan does not require additional power to cool the engine thereby saving fuel.

Keeping in mind the restrictions imposed to this analysis due to the limited engine geometry (absence of Radiator in the engine) available through the CAD model, a solution was developed to achieve this. A motivation behind this idea is the logic that the fan needs to do the extra work for cooling only when the coolant is not capable of providing the appropriate cooling.

The idea is to monitor the temperature of air flowing through the engine bay. Ideally, in a case where the air flow is not sufficient enough to cool the engine, the heat rate (or energy) of the air will be lower than that of the coolant (i.e., in simple terms, the air is not capable of helping the coolant in cooling function and the fan needs to run to perform the cooling function) and for cases where the airflow is sufficient, the heat rate of the air will be higher than that of the coolant. The idea will be more from Equation (1).

$$\text{Heat rate at the engine, } Q_{\text{gen}} = m C_p \Delta T \quad (1)$$

$$C_p = 0.862 \text{ kJ/kgK}$$

$$\Delta T = 366 - 358 \text{ K}$$

$$m = 4.115 \text{ kg/s}^{[3]}$$

$$Q_{\text{gen}} = m C_p \Delta T = 28.38 \text{ kJ/s}$$

Where m is the mass flow rate of the coolant, ΔT is the change in temperature of the coolant and C_p is the specific heat capacity of the coolant (ethylene glycol concentrate). The values shown here are reasonable values obtained from the website cited above and from other sources.

From the earlier argument, if $Q_{air} > Q_{gen}$, then the airflow is enough to cool the engine. If $Q_{gen} > Q_{air}$, then the incoming air cannot cool the engine and there will be fan-on condition.

This logic is implemented in such a manner that the temperature of air at the inlet (of the engine bay) and the outlet (just underneath the vehicle immediately after air leaves the engine bay) is monitored and the mass flow rate of air at the outlet is also monitored to prove that the air flow is enough to cool the engine.

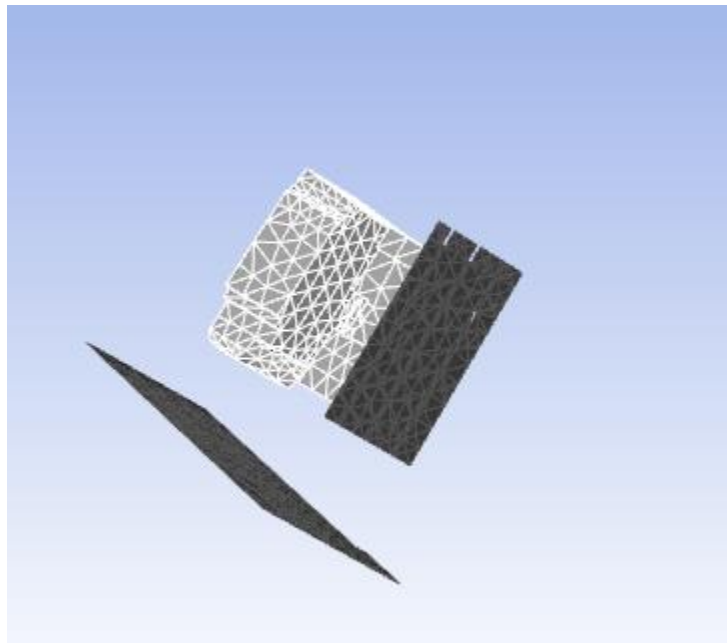


Fig 3.6: The planes to monitor temperatures

The above figure shows the way the above idea is being implemented. It can be observed that there are 2 planes created to measure the required values: the engine inlet plane and the outlet plane which are being used to monitor the inlet and outlet temperatures along with keeping an eye on the mass flow rate. This is because the basic idea here is heat transfer and that of a heat exchanger, the outlet mass flow rate should remain constant in this case (there is also possibility of deviations from this trend for a particular amount of time because of minor errors in setting the engine inlet plane) whereas the outlet temperature should increase appropriately in a manner indicative of heat transfer from the coolant to the air.

Assumptions:

The assumptions regarding the coolant has been mentioned above in the section on detailed study. The coolant assumed here is an ethylene glycol concentrate and the above C_p value is the specific heat values of the coolant. And the temperatures of the engine (ΔT) are typical values of engine temperatures. The m shown in the equation is the mass flow rate of the coolant in kg/s and it is similar to the flow rate of a Cummins ISX15 450 engine which is the engine used in the experiment here. Also, there is limited information on the Ambient conditions especially the wind direction. So, the wind speed here is assumed to be in the same direction and has a negligible effect on the vehicle speed. Another assumption here is that the key geometric features here defining the fluid flow consists of the cab (hood, grill, fenders, etc.); trailer incorporating box, side-skirts; undercarriage with wheels, frames. These features exist on a larger scale and therefore, their influence on aerodynamic forces overshadows those of small features such as mirrors, exhaust pipes, steps, etc. Therefore, these features are not considered for this project.

Another assumption that should be mentioned here is that the inlet temperature at the engine bay inlet is the same as the air temperature which I have assumed to be 300K. An argument point here will be whether the temperature air coming out of the engine bay of the first truck will affect the inlet air temperature at the second truck. Here, although the outlet temperature at the engine bay normally has an impact on the following vehicle, in our case it is not applicable as the outlet air temperature returns back to the original air temperature by the time it approaches the engine bay inlet of the second vehicle. The idea here is that the outlet temperature will be high at the plane of measurement and it gets cooled down by the ambient air once it crosses that plane.

Limitations and Challenges:

The first thing that you must consider when performing any sort of analysis is the limitations that you have. The first and most important limitation here was the availability of only the Academic Version of Ansys which limited the number of elements (or nodes) to 512,000. This limitation virtually eliminates the usage of such a big geometry as this can provide inaccurate meshes and cause significantly high and unnecessary computation costs. As a result of such a complex geometry which consisted of more than 700 faces overall (after using the half section) and complex structures, meshing the right portion is extremely significant. In order to keep the node and element number within the academic limits, a somewhat coarse mesh was utilized for this which has in turn affected some of those results, especially those which are susceptible to rapid change. The figure

below shows the mesh that was used for this simulation. In order to get a good enough mesh for the study, some body parts which can be avoided are suppressed.

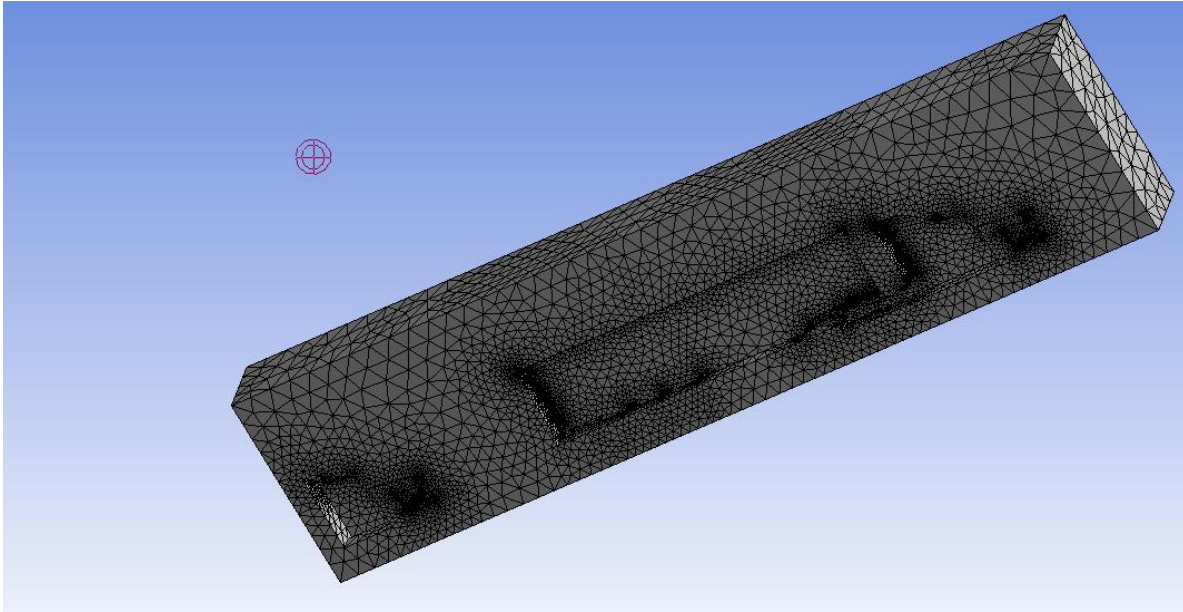


Figure 3.7: The meshed platoon formation inside the fluid domain

Another limitation was the unavailability of certain parameters which are important in this analysis such as the information about the coolant and the wind direction. Because of the unavailability of these parameters, these values were assumed for this project from online sources which has been mentioned before.

Another problem was the unavailability of the Radiator with the engine CAD model. The engine model used here is just a box enclosed inside the truck which means that the entire engine ‘mack’ here is to be defined as a constant temperature body which is technically inaccurate. Fig 3.7 below shows the picture of the so-called ‘box’ engine which has to be assigned a temperature value for this analysis. For this project, the box was therefore assumed to be a constant temperature and the concept of heat exchange was used to find out the temperature of the fluid flowing over it to simulate airflow through the engine bay.

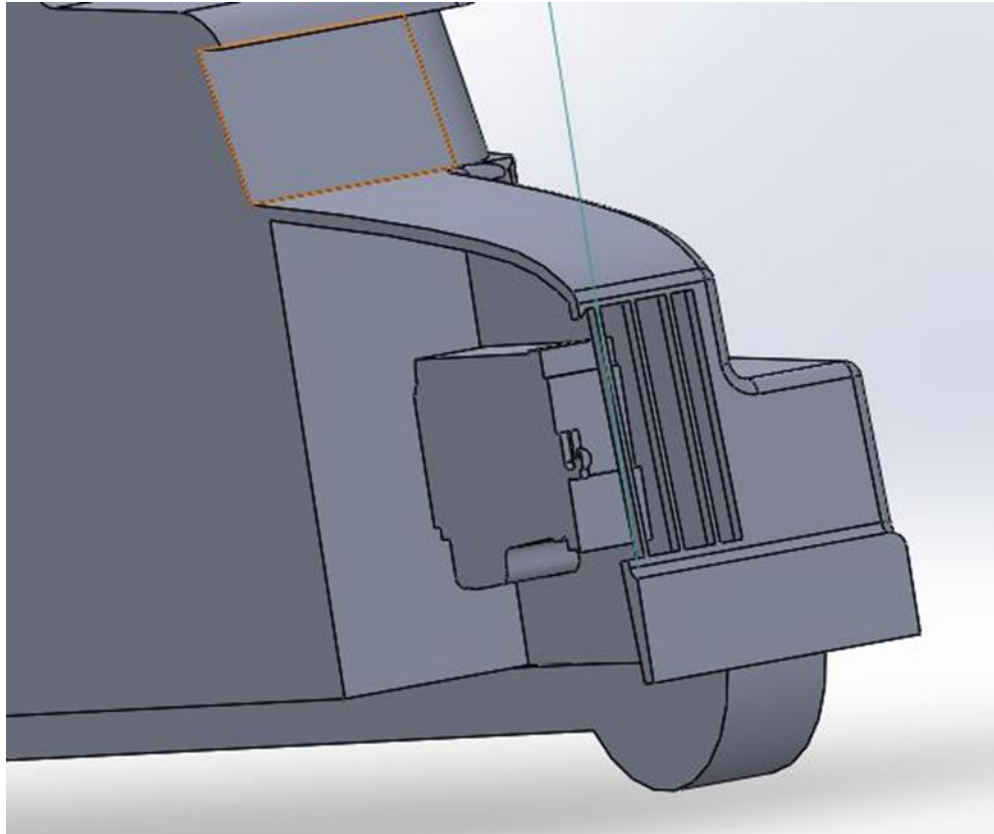


Fig 3.8: The Engine Mack model shown just as a box

Another major problem that was observed was pertaining to scaling. When scaling a model, in order to achieve equivalence with the full-scale model, there should be equivalence in the Reynold's number of both the full scale and the scaled model. In order to achieve such an equivalence in a testing media, if you are scaling down your model lengthwise by a factor of 0.01 then to get Reynold's number equivalence you must scale up your testing velocity by 100. In this case here, this means that the velocity of the scaled model had to be made 6500 mph (Mach-8) which was absurd. And since we are monitoring the mass flow rate also, it is also susceptible to such scaling errors and needs to be adjusted appropriately.

Shortcomings with the Experiment:

There are certain shortcomings with the experiment that needs to be pointed out. These so-called drawbacks greatly affect the capabilities of the analysis done in this project. One of the major information that is left out in this paper is the information on the wind direction which can greatly impact the CFD wind tunnel velocity. The only information present in the paper is the windspeed which is not useful unless the wind direction is accurately specified (i.e., whether the wind is in

the same direction or flowing against the vehicle or if there is a cross-wind). Here for this experiment, the wind was assumed to be in the same direction as there are certain cryptic indications in the paper that may suggest that wind was in the same direction though there is no significant support to this assumption.

Another major issue that suggest that this experiment requires validation is the lack of necessary information on the Radiator and the coolant. There is a table which shows the percentage increase or decrease in coolant temperature or the air temperature as recorded by the CAN bus. This information can only be useful if backed by accurate numbers indicating the increase from a certain value. An accurate representation of coolant information could have provided some solid validation comparison with this project.

Alternative study manners:

There are other ways of conducting this study which are not actually discussed as part of this project, but which are worth mentioning. A popular way is using the methodology of forced convection using heat flux and the heat transfer coefficient in order to find out which iteration has produced the lowest or highest values for the change in temperature.

Another way is by modifying the engine design to include a Radiator which virtually makes it easier to conduct the CFD as it allows for a much more realistic assessment and eliminates the meshing issues.

CHAPTER 4

RESULTS

This section discusses the results of the CFD study done in this project. As mentioned before, to overcome the challenges that were discussed above and to save computation cost, an equivalence study was conducted, and the results were produced with an equivalent structure which is similar to the real model. The results shown below indicate that the airflow through the engine bay is enough to cool the engine for higher following distances and prevent the engine fan from drawing power to run and thereby reducing fuel efficiency. The preliminary study Figures 3.4 and 3.5 has already shown that the air-particle flow through the engine bay is better during higher following distances when the air-flow is approaching steady state in comparison with lower following distances.

As mentioned before, from the experiment it is evident that the best fuel efficiency condition is obtained when the Gross Vehicle Weight is 65,000 lbs. Therefore, the results shown here are only for this weight condition. The figure below shows the inlet temperature over different iterations. As expected, the inlet temperature is staying constant. The temperature here is the ambient air temperature which has been assumed as 300K with the information from the paper. This value is the same for all the results irrespective of the following distances.

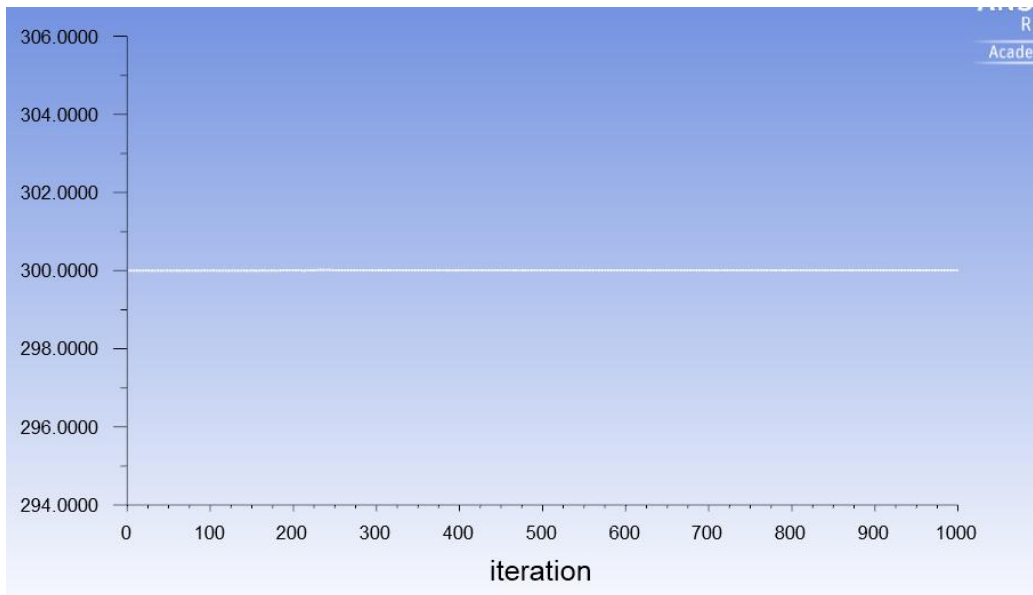


Fig 4.1: Constant inlet temperature of the fluid (air)

These results are shown for the both the full-scale model and one of the scaled models but in order to avoid the computation cost only a half section of the tractor-trailer platoon is being considered as in Fig 3.2. In order to save computational cost and due to limited time availability in the lab, only 1000 iterations were run for each condition. Now, the results for outlet temperature for the various following distances are given below:

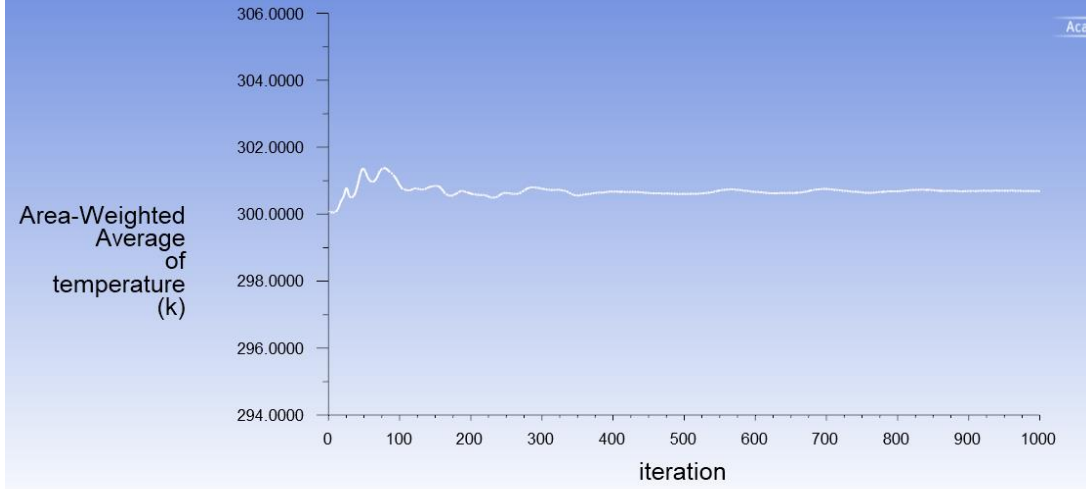


Fig 4.2: Outlet temperature when the following distance is 20ft

The above figure shows the outlet temperature at the outlet plane (as seen from fig 3.6). Here the following distance is 20ft which means the trailing vehicle is closely following the lead vehicle. So, as evident from the preliminary study, lesser air will be flowing through the engine bay. And, this air will get the heat from the coolant and as a result, the outlet air will have higher temperature. But here we can see from Fig 4.2 that the air temperature is being increased only marginally to around 301.5K which means from Equation (1) that $Q_{gen} > Q_{air}$. This case will become clearer as we discuss the other cases. An interesting observation which was not expected during the background study is that the mass flow is not staying constant. But since the temperature is also showing a downward trend slightly in fig 4.2, such a fluctuation is justified. Either way, the Q values remains the same after some initial iterations.

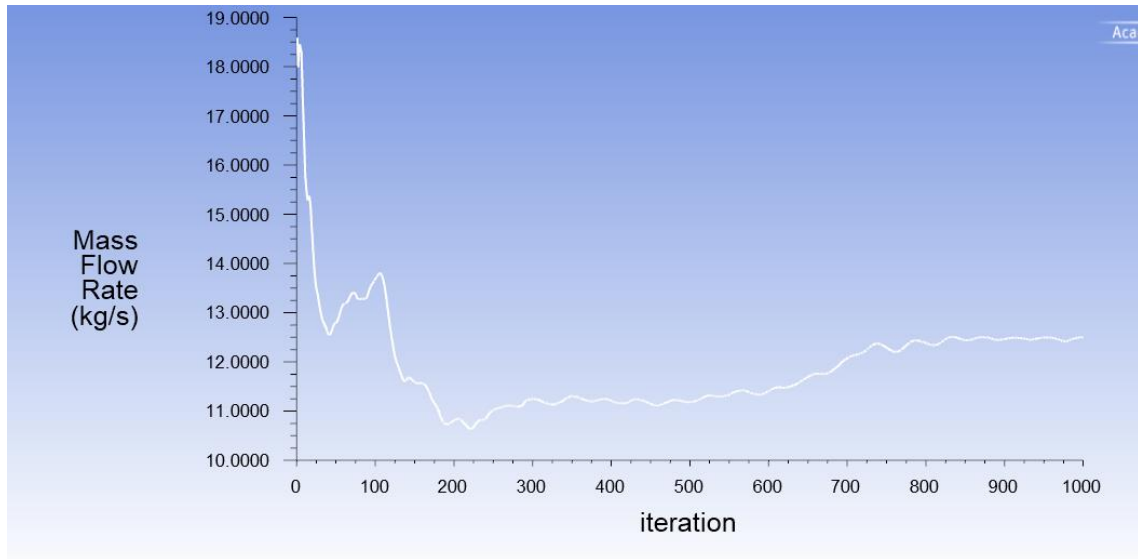


Fig 4.3: Outlet mass flow rate for 20ft

From the above figure, it can also be seen that the mass flow rate is slowly approaching a constant rate with each iteration as the temperature plot in Fig 4.2 is fast approaching a constant value. Once the air passes the monitoring plane, it can be assumed that the air will get cooled down by the ambient conditions and as a result, the final air coming from the back of the trailer will have the same temperature as the surrounding air as mentioned before.

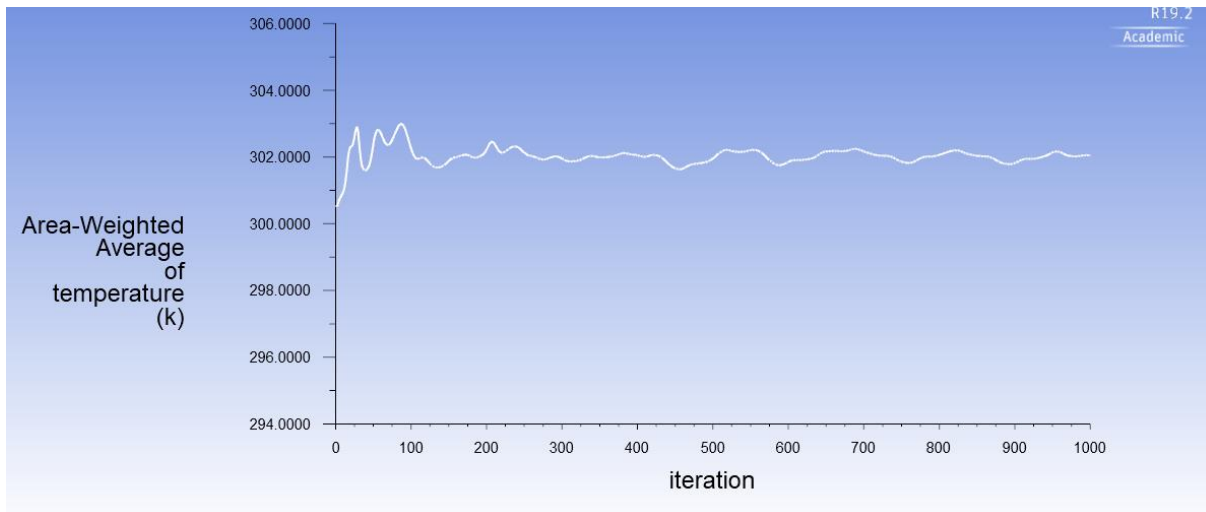


Fig 4.4: Outlet temperature when the following distance is 70ft

The above figure shows the outlet temperature when the following distance between the lead tractor and the trailing tractor is 70ft. The picture seen here is for the scaled model. The velocity was scaled up to get Reynold's number equivalence. Here, the temperature is increasing to a higher value (303K) in comparison with fig 4.2. This implies that the air is getting more heated up in this

case than when the following distance was 20ft. This means that there is more heat exchange between the coolant and the incoming air resulting in reduced fan usage for cooling. This proves that the airflow is higher when the following distance is higher, the air can sufficiently cool the engine to reduce the power consumption due to the fan usage. Similar to the previous scenarios, here also the outlet temperature is approaching a constant value with increasing iterations.

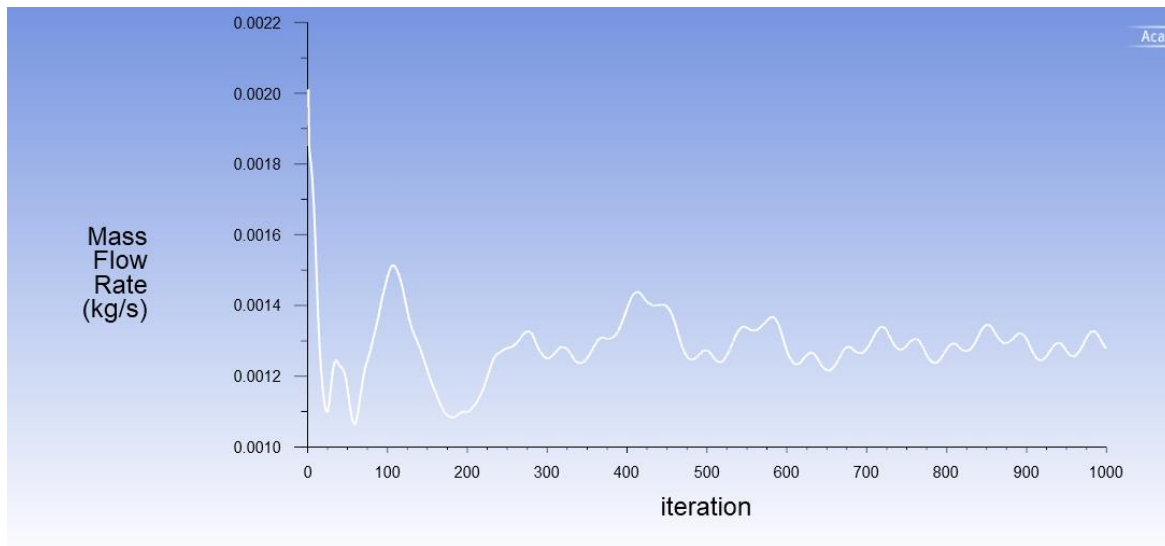


Fig 4.5: Mass flow rate for 70ft following distance

The above figure shows the corresponding mass flow rate for 70ft following distance. Notice that the mass flow rate needs to be appropriately adjusted as we're using a scaled model (multiplying by 100). Similar to the above scenarios also, the mass flow rate here is also slowly converging to a constant value while the temperature value has almost approached a constant value. But, despite all these, the Q value is believed to stay after some iterations.

The simulations with 30ft and 40ft following distance also showed similar values for outlet temperature as the values for 20ft with only minor variations. Therefore, these results are not included here since they don't form effective grounds for comparison.

And as expected the simulation for 50ft showed a somewhat intermediate value between those for 20ft and 70ft. The figure below shows the simulation results for outlet temperature when the following distance between the lead and the following vehicle is 50ft.

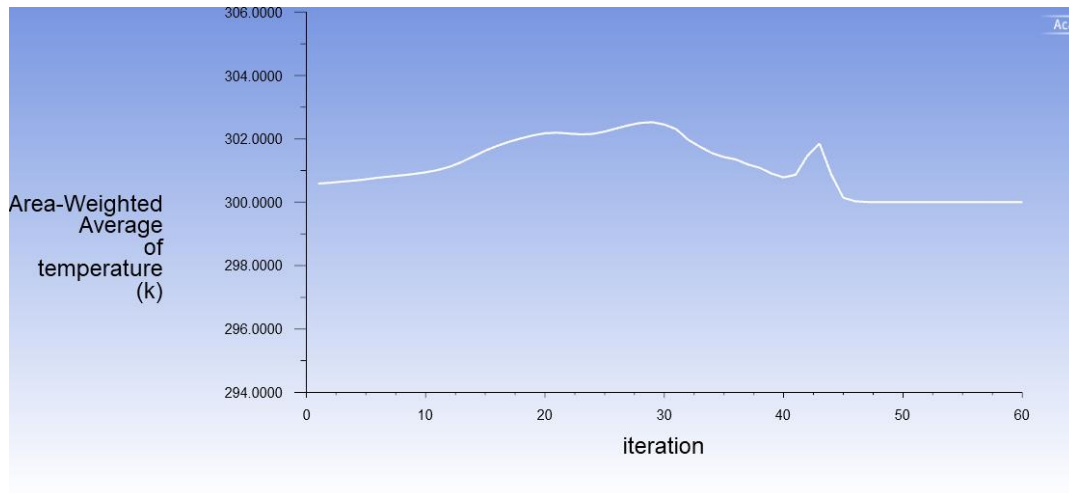


Fig 4.6: Outlet temperature when the following distance is 50ft

The above figure shows that the outlet temperature reaches a value of just above 302K which lies somewhere in between that of 20ft and 70ft. It is also evident that the value approaches a constant value same as the ambient after some time.

All these results indicate that when the following distance is 70ft, the airflow is high enough (condition where $Q_{air} > Q_{gen}$ as explained before) that the coolant is enough for the engine cooling and the engine fan need not perform extra work and cause additional fuel consumption.

As explained before and as is evident from these figures, the fuel efficiency losses (thermal loss) due to the reduced air flow through the engine bay is very minimal. Further analysis can reveal an optimum distance where both of these losses are optimum.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

The CFD results validate the experimental results (shown in Appendix A). But there are certain uncertainties that need to be addressed in the experiment as mentioned before. Basically, at higher following distances, the air flowing through the engine bay will be sufficient enough to cool the engine and reduce the fan-on conditions thus reducing the power consumption. When the trailing vehicle is following closely to the lead vehicle, the aerodynamic drag is going to be reduced. However, the air following through the engine bay is going to be reduced and the fan on condition is going to increase power consumption in these cases.

Recommendations:

Conducting the experiment once again using a different driving schedule and by studying the impacts to the airflow through the engine bay when there is another vehicle in between the lead and the trailing vehicle just like in a real-life environment. Also, testing by considering only the Longitudinal motion without considering the Lateral and Yaw motion seems to be a bit unrealistic and doesn't seem to be a convincing cause for judging the fuel efficiency benefits of a platoon formation. So, as is also evident from point 4 of the Chapter 2, ^[8] a better test in a simulated test environment with traffic and considering cornering and gradient changes needs to be conducted to get a better picture of fuel efficiency benefits of platooning and whether aerodynamic losses overrule the thermal losses.

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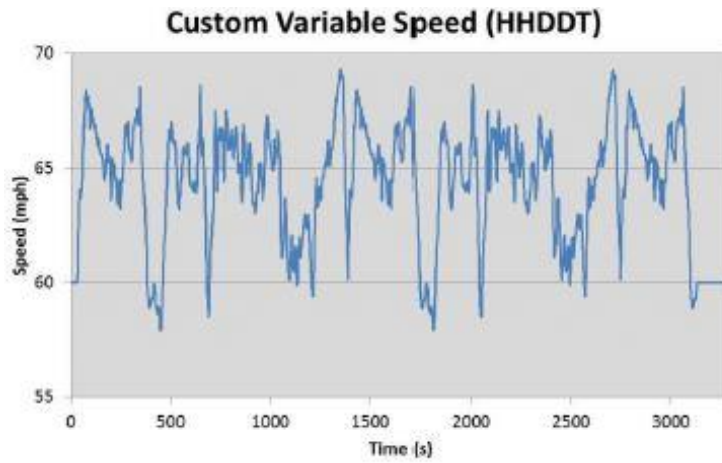
APPENDIX A ^[1]

EXPERIMENT INFORMATION

Experiment Tractor-Trailer Specifications:

Van Specification	Lead Tractor-Trailer	Trailing Tractor-Trailer	Control Tractor-Trailer
Manufacturer	Peterbilt	Peterbilt	Peterbilt
Model	386	386	579
Model year	2011	2011	2013
Vehicle Mileage at Test Start	325,367	360,334	43,618
Engine manufacturer	Cummins	Cummins	PACCAR
Engine model	ISX-15 450	ISX-15 450	MX-13 475
Engine model year	2010	2010	2010
Emissions equipment	DOC/DPF/SCR	DOC/DPF/SCR	DOC/DPF/SCR
Transmission	Eaton Fuller 10-speed	Eaton Fuller 10-speed	Eaton Fuller 10-speed
Retarder/regenerative braking	Engine Brake	Engine Brake	Engine Brake
Tires	Michelin XDA Energy	Michelin XDA Energy	Goodyear Fuel MAX
Cab Fairing to nose of trailer (visible space between tractor and trailer)	50"	50"	41"
Trailer Configuration	53-ft Van Trailer with angled side skirts	53-ft Van Trailer with angled side skirts	53-ft Van Trailer with angled side skirts
Trailer Manufacturer	Great Dane	Stoughton	Wabash
Trailer Model	74111 SSLAK	2GPVW-53ST-S-L-AR	TRA/ VAN
Trailer Height	13' 6"	13' 6"	13' 6"
Trailer Width	102"	102"	102"
King Pin Set Back	36"	36"	36"
Trailer Side Skirt	Freight Wing AeroFlex	Freight Wing AeroFlex	DuraPlate AeroSkirt
Distance from rear of trailer side skirt to front of trailer wheels	6.5"	8.5"	9.5"

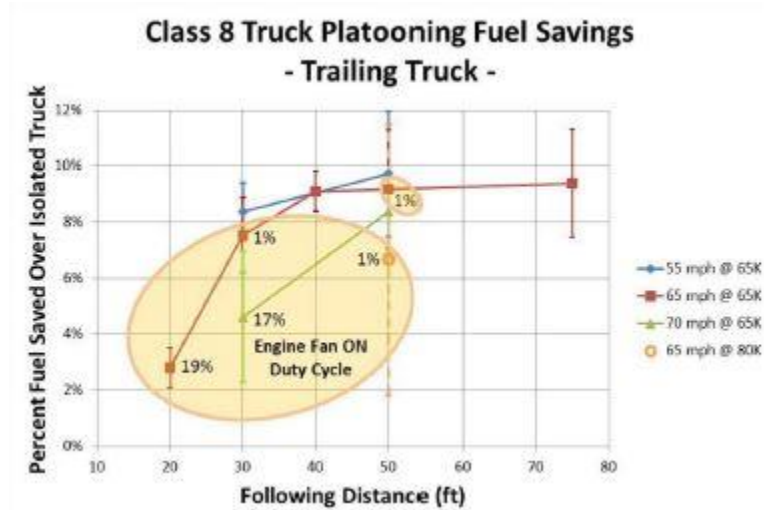
Speed Trace based on HHDDT:



Percentage Fuel saved of Trailing Tractor:

<u>Test Conditions</u>	Trailing Truck	
	<u>Nominal</u>	<u>95% CI</u>
55 mph, 65k, 30 ft	8.38%	1.03%
55 mph, 65k, 50 ft	9.72%	2.24%
65 mph, 65k, 20 ft	2.81%	0.71%
65 mph, 65k, 30 ft	7.53%	1.33%
65 mph, 65k, 40 ft	9.10%	0.71%
65 mph, 65k, 50 ft	9.17%	2.11%
65 mph, 65k, 75 ft	9.39%	1.93%
70 mph, 65k, 30 ft	4.62%	2.33%
70 mph, 65k, 50 ft	8.36%	1.44%
VAR, 65k, 50 ft	4.22%	6.82%
65 mph, 80k, 50 ft	6.67%	4.84%

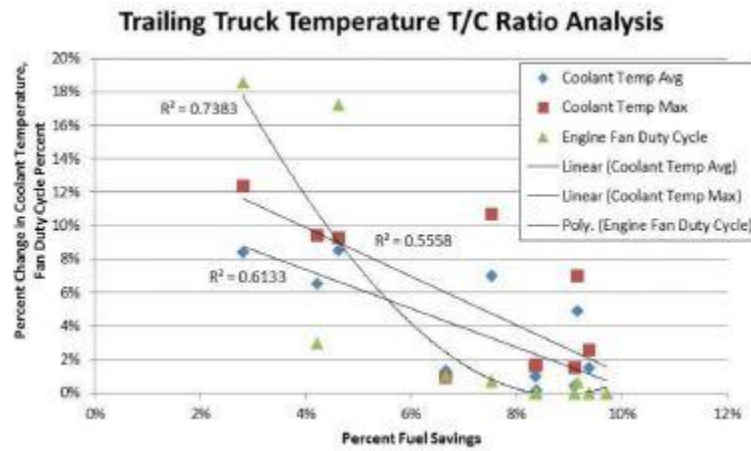
Trailing Platoon Tractor Fuel Savings:



Average and Maximum Coolant and Intake air temperature changes observed on the Trailing Tractor:

Test Conditions	Fuel Savings	Coolant Temp Avg	Coolant Temp Max	Intake Air Temp Avg	Intake Air Temp Max	Engine Fan Duty Cycle
65 mph, 65k, 75 ft	9.4%	1.5%	2.6%	0.2%	-4.4%	0%
65 mph, 65k, 50 ft	9.2%	4.9%	7.0%	4.6%	9.0%	1%
65 mph, 65k, 40 ft	9.1%	0.4%	1.5%	6.7%	2.3%	0%
Test Conditions	Fuel Savings	Coolant Temp Avg	Coolant Temp Max	Intake Air Temp Avg	Intake Air Temp Max	Engine Fan Duty Cycle
65 mph, 65k, 30 ft	7.5%	7.0%	10.7%	0.8%	11.8%	1%
65 mph, 65k, 20 ft	2.8%	8.4%	12.4%	-2.4%	13.2%	19%
55 mph, 65k, 50 ft	9.7%	-0.3%	-0.6%	-22.0%	-14.0%	0%
55 mph, 65k, 30 ft	8.4%	0.2%	1.7%	-9.5%	0.4%	0%
70 mph, 65k, 50 ft	8.4%	1.0%	1.6%	5.8%	1.7%	0%
70 mph, 65k, 30 ft	4.6%	8.5%	9.3%	24.3%	22.4%	17%
VAR, 65k, 50 ft	4.2%	6.6%	9.4%	0.9%	9.6%	3%
65 mph, 80K, 50 ft	6.7%	1.3%	0.9%	-10.8%	-4.1%	1%

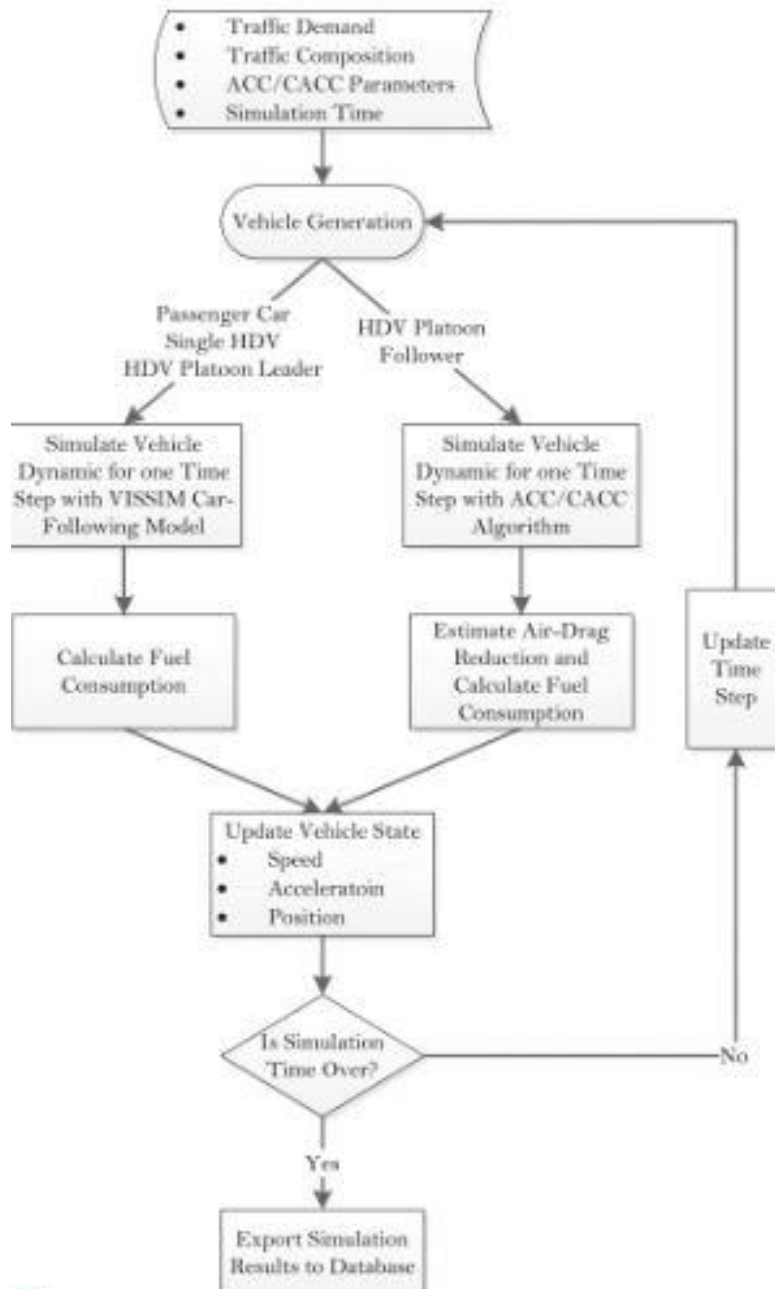
Trailing Tractor Cooling Temperature Analysis:



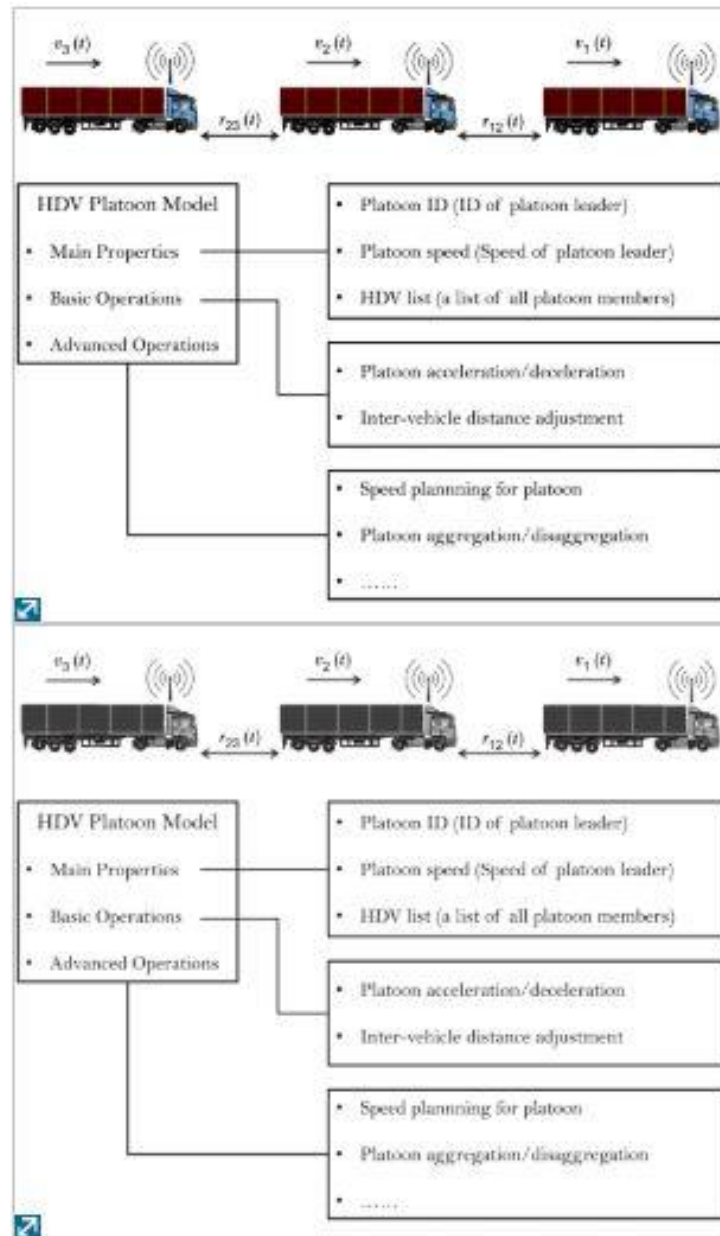
APPENDIX B

PLATOON SIMULATION AND ANALYSIS

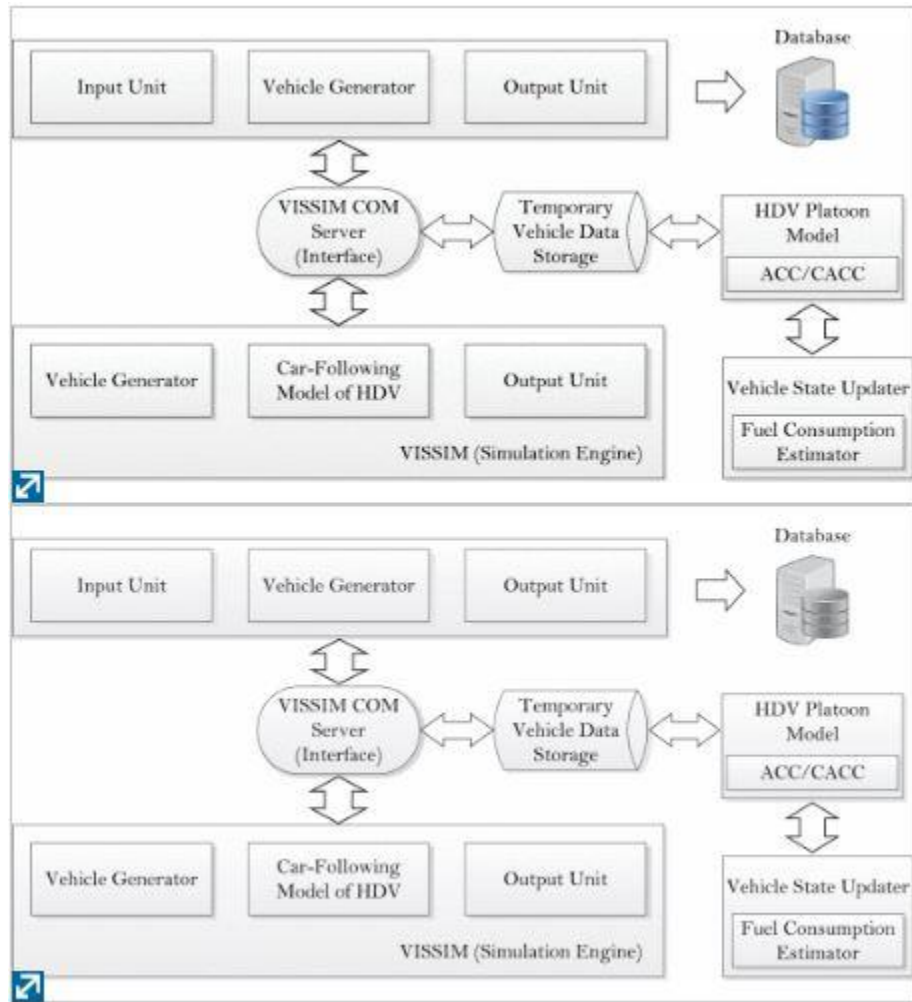
Simulation Process Flowchart: ^[8]



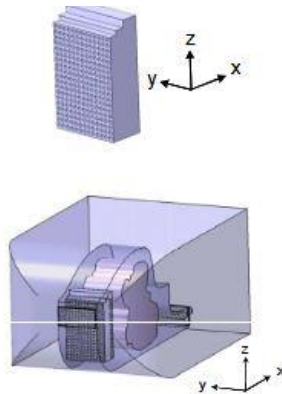
HDV Platoon Model with its properties and basic and advanced operations: ^[8]



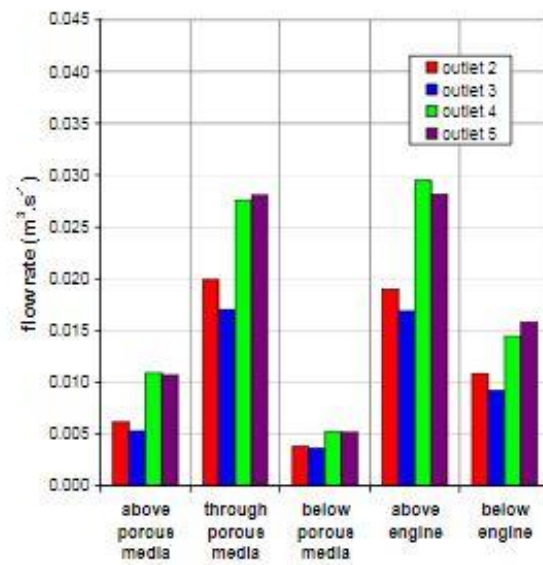
Simulation Framework Architecture: [8]



Porous Media and Simplified Engine Geometry in isometric view: [2]



Flow rates in the engine compartment: [2]



Total vehicles moving at a given point throughout the 24-hour period: [4]

